

Essential Fatty Acid (EFA) Deficiency

Affected individuals may also demonstrate hyperpigmentation or hypopigmentation of the hair. Fatty liver infiltration, increased susceptibility to infection, a blunted immune response, anemia, thrombocytopenia, and growth retardation can also be observed.

Laboratory Testing

In EFA deficiency states, linoleic acid levels are low. The enzymes that normally convert linoleic acid to arachidonic acid instead use oleic acid, creating the abnormal byproduct 5,8,11-eicosatrienoic acid. Laboratory evaluation therefore shows:

- Decreased levels of linoleic acid and arachidonic acid
- Elevated plasma levels of 5,8,11-eicosatrienoic acid

An increased ratio (≥ 0.2) of 5,8,11-eicosatrienoic acid to arachidonic acid is diagnostic of EFA deficiency.

Treatment

Topical application of sunflower seed and safflower oils—rich in linoleic acid—may improve cutaneous manifestations of EFA deficiency. However, topical absorption is unpredictable. Optimal treatment typically involves oral or intravenous essential fatty acid supplementation. To prevent deficiency, EFAs should constitute 1% to 2% of total daily caloric intake.

Micronutrients

Fat-Soluble Vitamins

Vitamin A (Retinol)

Etiology and Pathogenesis

Vitamin A is a fat-soluble vitamin essential for retinal photoreceptor function, epithelial cell proliferation, and keratinization. The two most clinically important metabolites are:

- **Retinal:** a key component in rhodopsin synthesis
- **Retinoic acid:** regulates cell differentiation

Dietary sources of vitamin A include:

- **Plant sources:** dark green leafy vegetables, red palm oil, and brightly colored fruits (e.g., papaya, mango, carrots, tomatoes, apricots, cantaloupe). In plants, vitamin A is present as β -carotene, which is cleaved to form retinal. Retinal is then reduced to retinol in intestinal villous cells.
- **Animal sources:** egg yolk, liver, fish, fortified milk, and dairy products. In animals, vitamin A exists as retinyl esters, which are hydrolyzed to retinol in the intestinal lumen and absorbed by mucosal cells.

All absorbed retinol is re-esterified to retinyl esters in the intestinal mucosa, packaged into chylomicrons, and transported to the liver for storage. When needed, hepatic retinyl esters are converted back to retinol, bound to retinol-binding protein (RBP) and transthyretin, and released into circulation.

Vitamin A Deficiency (VAD)

Epidemiology

Vitamin A deficiency (VAD) can lead to both cutaneous and ocular complications. It is the most common cause of preventable childhood blindness worldwide, according to the WHO. VAD is also associated with impaired immune function.

Etiology and Pathogenesis

Primary causes of VAD include:

- Inadequate dietary intake
- Fat malabsorption syndromes
- Liver disease

In the United States, risk factors include eating disorders, restrictive diets, and chronic illness. Due to its fat-soluble nature, vitamin A absorption is impaired in conditions such as:

- Pancreatic or biliary tract disease
- Celiac disease
- Crohn disease
- Shwachman-Diamond syndrome
- Cystic fibrosis
- Cholestatic liver disease
- Gastric bypass surgery

Clinical Findings

Ocular Manifestations (earliest signs):

- Impaired dark adaptation (**nyctalopia**)
- **Xerophthalmia** (dry eyes)

- **Bitot spots** – white, foamy patches on the conjunctiva due to keratin desquamation and bacterial overgrowth
- **Corneal xerosis**, ulceration, and **keratomalacia** (softening of the cornea), which may progress to corneal perforation, iris prolapse, and irreversible blindness

Cutaneous Manifestations:

Result from abnormal keratinization:

- **Mild deficiency**: dry skin (xerosis) and scaling
- **Severe deficiency**: deep skin fissures, known as **dermomalacia**